

Overview of the math-proofs Collection

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Abstract

This document describes the result families in the `math-proofs` collection: machine-checked mathematics in which every proven result ships as a paper preprint (human-readable argument) paired with its Lean 4 formalization (machine-checked proof). Manuscripts are organized into families; each family groups a principal result with companion arguments, consequences, or alternative proofs. All formalizations typecheck with `lake build` under Lean 4.34.1 + Mathlib v4.34.1. The per-theorem verification status and axiom dependencies are recorded in `formalization.yaml` (machine-readable) and `FORMALIZATION_CATALOGUE.md` (human-readable).

1 Labeling convention

Label	Meaning
[PROVED]	Machine-checked by Lean; no axioms beyond Lean’s standard ones.
[PROVED*]	Typechecks cleanly but depends on explicitly labeled classical-input axioms.
[SOLVED]	Computationally established on independent code paths (no Lean proof yet).

Table 1: Status labels used throughout the collection.

Nothing in this collection is claimed on the basis of a `sorry`.

2 Family 001 — Erdős–Straus identities and obstructions

Results on representing $4/n$ as a sum of three unit fractions.

ID	Result	Status	Lean module
ES-001	New covering identity for $n \equiv 217 \pmod{264}$	[PROVED]	<code>ESIdentity.lean</code>
ES-002	Finite-identity barrier on square classes mod 840	[PROVED*]	<code>ESBarrier.lean</code>

Table 2: Family 001 manuscripts.

ES-001 gives a new covering identity

$$\frac{4}{n} = \frac{1}{(n+3)/4} + \frac{1}{n(n+3)} + \frac{11}{n(n+3)}, \quad n \equiv 217 \pmod{264},$$

covering the prime 1009 — the smallest prime missed by the published Mordell/Yamamoto/Rosati identities. The formalized theorem is `es_identity_217mod264`, depending only on Lean’s standard axioms (`propext`, `Classical.choice`, `Quot.sound`).

ES-002 proves that no finite system of polynomial identities $4/(A_it + B_i) = \sum 1/F_{ij}(t)$ with positive leading coefficients can cover a full square class mod 840: infinitely many primes escape every such system. Witness: $p = 77,598,049 \equiv 529 \pmod{840}$. The formalization carries two explicitly labeled classical-input axioms: `schinzel_mordell` and `dirichlet_arith_prog`.

3 Family 002 — Collatz convergent control (gap-factor theory)

A machine-checked theory of the gap factor $G(t)$ on from-above blocks of $\log_2 3$'s continued fraction, developed across thirteen Lean modules: the complete-quotient API for irrationals (`CollatzGapCF`), the C24 tail-mirror selector and strict dichotomy (`CollatzGapC24`, `CollatzGapC24Lemma3`), C23 sinh estimates (`CollatzGapC23`, `CollatzGapSinh`), the C22/C25/C26 assembly (`CollatzGapC22C25`), C18b convexity and the C26 margin-duality trichotomy (`CollatzGapConvex`), C29 discrepancy dominance and its from-below mirror (`CollatzGapC29`, `CollatzGapC29Mirror`), and the C64/C64b hump-law identity kernel (`CollatzGapHump`). All modules are [PROVED] with standard axioms only; preprints are in preparation.

4 Family 003 — Erdős–Straus witness-complex and QR-core theory

A machine-checked theory of Erdős–Straus covering families on refined moduli: the witness complex $W(M)$ (`WitnessComplexThm2`, `WitnessComplexThm13`), the T2 prime-square obstruction (`ESPrimeSquareT2`), the T1 two-case theorem (`EST1TwoCase`), and the QR-core — the set of quadratic-residue classes missed by both covering systems — with its exact count $|Q| = |S|/2^k$ (`ESQRCore`, `ESQRCoreCount`), the 5/7-forcing lemma (`ESFiveSevenForcing`), the mod-7 universal families and fiber-pinning lemma (`ESMod7UniversalFamilies`), the X0/S1S3 descent core (`ESXOS1S3`), general descent transfer (`ESDescentTransfer`) and the pair-specific conditional SDR endgame (`ESBranchSDREndgame`). All modules are [PROVED] with standard axioms only; preprints are in preparation.

5 Family 004 — Girth-5 extremal graphs

G5-001 is the 3v floor theorem for girth-5 graphs (`Girth5Floor3v.lean`, [PROVED]): the emitted-graph re-attachment model and cross-edge harmlessness argument, showing that no improving 3v move exists on the extremal graphs. Preprint in preparation.

6 Family 005 — q-adic classification of Erdős–Straus witnesses

QA-001 is the branch-1 q-adic type theorem (`ESQadicBranch1.lean`, [PROVED]): $v_q(r) \leq v_q(A)$ on refined targets, the first classification theorem for the q-adic types of Erdős–Straus witnesses. Preprint in preparation.

7 Family 006 — Saturated Sidon sets

SS-001 ([SOLVED]): the set $\{16, 62, 66, 74, 88, 104, 117, 122, 123\}$ is Sidon and saturated in $[1, 184]$, verified on three independent code paths. Paper: `papers/r47-sat184-October-7-2026/`.

Family	Subject	Modules
001	Erdős–Straus identities and obstructions	2
002	Collatz convergent control (gap-factor theory)	13
003	Erdős–Straus witness-complex and QR-core theory	11
004	Girth-5 extremal graphs	1
005	q-adic classification of ES witnesses	1
006	Saturated Sidon sets	0 (computational)

Table 3: The six result families and their formalization footprint.

8 Summary of families

Versions and citations

Corrections and revisions are recorded as new versions in `CHANGELOG.md`, with previously released versions remaining accessible in git history. To cite an individual manuscript, use the BibTeX block in its `papers/<slug>/CITATION.bib`.